

HOW DOES YOUR BRAIN STORE MEMORIES?

~86 BILLION NEURONS

SYNAPSE: ~TRILLIONS OF CONNECTIONS

MEMORY = A PATTERN, NOT A FILE

NEURONS THAT FIRE TOGETHER, WIRE TOGETHER

LTP: PRACTICE STRENGTHENS SYNAPSES

SENSORY MEMORY: < 1 SECOND

**WORKING MEMORY: ~7 ITEMS, ~15–30
SEC**

LONG-TERM MEMORY: NEARLY UNLIMITED

HIPPOCAMPUS: BUILDS NEW MEMORIES (CONSOLIDATION)

SLEEP STRENGTHENS MEMORIES

MUSCLE MEMORY (CEREBELLUM)

UNUSED = FADES

**QUICK BRAIN CHECK! — PAUSE &
ANSWER\n1) How many neurons? 2)
Name the gap? 3) Which part builds
long-term memories?**

**ANSWERS:\n1) ~86 BILLION NEURONS
2) SYNAPSE 3) HIPPOCAMPUS**

**RECAP: NEURONS • SYNAPSES • LTP •
HIPPOCAMPUS • SLEEP\nSUBSCRIBE &
SHARE!**